Noise-Resilient Time-Series Forecasting via Generative Residual Learning

Chathurika S Abeykoon
abeykoonc@rhodes.edu

Aleem Azhar
azhmu-26@rhodes.edu

Golam Hisham
hisgo-26@rhodes.edu

authorD
authorD@email.com

Abstract

Time-series forecasting models often struggle when observations contain complex stochastic noise. Traditional regression approaches focus on learning deterministic temporal relationships but may fail to capture nonlinear noise structures present in real-world data such as financial markets.

This paper proposes a hybrid framework called Generative Residual Learning, which separates forecasting into two complementary tasks: learning the deterministic signal using regression and learning the stochastic residual distribution using a generative model. By modeling residual noise explicitly, the proposed approach improves robustness to noisy observations while preserving interpretability of regression-based forecasting.

We provide theoretical insight showing that the expected prediction error of the hybrid model is bounded by the regression error and the distance between true and learned residual distributions. Empirical results on simulated time-series data demonstrate that the proposed approach produces more stable forecasts and lower mean squared error compared to classical regression models.

Keywords: time-series forecasting, generative modeling, residual learning, robust regression, noise modeling, financial forecasting, machine learning

1 Introduction

Time-series forecasting plays a central role in data-driven decision-making across a wide range of domains, including financial risk management, energy load balancing, healthcare monitoring, climate modeling, and supply chain optimization. In many of these settings, forecasts must be generated from high-dimensional, noisy, and partially observed data streams. The increasing availability of large-scale temporal datasets has motivated the development of sophisticated statistical and machine learning methods capable of capturing complex temporal dependencies.

Classical statistical approaches, such as Autoregressive Integrated Moving Average (ARIMA), exponential smoothing, and state-space models, provide interpretable and theoretically grounded frameworks for forecasting. These methods perform well when the underlying dynamics are approximately linear and stationary. However, real-world time series often exhibit nonlinear interactions, regime shifts, heteroskedasticity, long-range dependencies, and structural breaks. In such scenarios, linear parametric models may suffer from model misspecification, leading to biased forecasts and degraded performance.

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Deep learning architectures have emerged as powerful alternatives for modeling nonlinear temporal dynamics. Recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models, are designed to capture sequential dependencies through gated memory mechanisms. More recently, Transformer-based architectures have demonstrated strong performance by leveraging self-attention mechanisms to model long-range temporal interactions. While these approaches significantly improve predictive accuracy in many applications, they remain sensitive to noisy inputs, missing data, adversarial perturbations, and distributional shifts between training and deployment environments. Moreover, most deep forecasting models are trained purely through pointwise regression losses (e.g., mean squared error), which may fail to capture the full conditional distribution of future trajectories.

Generative Adversarial Networks (GANs), originally proposed for realistic data generation, provide a distributional learning framework based on a two-player minimax game between a generator and a discriminator. By optimizing divergence measures between generated and true data distributions, GANs have demonstrated remarkable ability to model complex, high-dimensional structures in images, text, and other modalities. In the context of time-series data, GAN-based approaches offer the potential to learn richer predictive distributions rather than merely point estimates. However, standard GAN formulations are unstable to train, prone to mode collapse, and not directly tailored for regression-based forecasting tasks.

2 Related Work. ** add at least 1-2 citations for each section below

2.1 Classical Time-Series Forecasting

Time-series forecasting has traditionally been studied within statistical modeling frameworks such as AR, MA, ARIMA, and state-space models. These methods assume linear dependence structures and often rely on stationarity or weak nonstationarity assumptions. Extensions including GARCH-type models address heteroskedasticity, while Kalman filtering enables latent state inference in linear Gaussian systems. Although these approaches provide interpretability and theoretical guarantees, their performance degrades when temporal dynamics are strongly nonlinear, high-dimensional, or subject to structural breaks. Moreover, their parametric assumptions limit flexibility in modeling complex real-world systems.

2.2 Deep Learning for Sequential Modeling

The success of deep learning has led to widespread adoption of neural architectures for forecasting. Recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models, are designed to capture nonlinear temporal dependencies via gated memory mechanisms. Sequence-to-sequence models further extend this framework for multi-step forecasting. More recently, Temporal Convolutional Networks (TCNs) and Transformer-based models leverage convolutional or attention mechanisms to capture long-range dependencies without strict recurrence.

Despite strong empirical performance, most deep forecasting models are trained using pointwise losses such as mean squared error (MSE) or mean absolute error (MAE). These objectives focus on conditional mean estimation and may fail to capture multimodality, heavy tails, or full predictive uncertainty. Furthermore, deep models can be sensitive to noisy inputs, missing values, and distributional shifts between training and testing environments, which are common in financial, medical, and sensor-based data.

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2.3 Probabilistic and Distributional Forecasting

To better model uncertainty, several probabilistic forecasting approaches have been proposed. Bayesian RNNs introduce parameter uncertainty through variational inference or Monte Carlo dropout. Deep state-space models combine latent variable structures with neural parameterizations. Variational autoencoder (VAE)-based time-series models learn generative representations of temporal dynamics. Quantile regression and distributional loss functions have also been used to estimate predictive intervals.

While these methods improve uncertainty quantification, variational approaches may rely on restrictive posterior approximations, and likelihood-based training can become intractable or unstable in high-dimensional settings. Moreover, explicit density modeling may require strong parametric assumptions that are difficult to justify in complex real-world systems.

2.4 Generative Adversarial Networks for Time Series

Generative Adversarial Networks (GANs) provide a likelihood-free alternative for distribution learning by optimizing a minimax objective between a generator and a discriminator. Wasserstein GANs (WGANs) improve training stability by replacing divergence-based objectives with the Earth Mover (Wasserstein-1) distance, providing smoother gradients and theoretical grounding in optimal transport.

Several works have adapted GANs for time-series generation and imputation, including RNN-based generators and critics for sequential data. GAN-based approaches have shown promise in synthetic time-series generation, anomaly detection, and data augmentation. However, most existing GAN models focus on unconditional or conditional sequence generation rather than supervised forecasting. When applied to prediction, adversarial losses alone may fail to ensure accurate alignment with ground-truth future values, leading to realistic but biased forecasts.

2.5 Hybrid Adversarial–Regression Approaches

Recent efforts have explored combining adversarial objectives with supervised losses in tasks such as image-to-image translation and structured prediction. In forecasting, hybrid models incorporate regression losses alongside adversarial training to balance realism and accuracy. However, these approaches are often heuristic, lack theoretical justification, or rely on standard GAN losses that are prone to instability and mode collapse.

2.6 Our Contribution

Our proposed Regression-WGAN (R-WGAN) builds upon Hybrid Adversarial–Regression line of work by integrating a Wasserstein adversarial objective with an explicit regression loss tailored for multi-step time-series forecasting. Our approach integrates adversarial distribution matching with an explicit regression objective, thereby combining the strengths of generative modeling and supervised learning. The Wasserstein formulation provides improved training stability and meaningful gradient signals, mitigating issues commonly observed in vanilla GANs. Simultaneously, the regression loss enforces alignment between generated trajectories and ground-truth future values, ensuring predictive accuracy alongside distributional realism.

A key motivation of our framework is robustness under noisy and corrupted inputs. In many real-world systems—financial markets with microstructure noise, sensor networks with measurement errors, medical time-series with missing values—observations deviate substantially from idealized assumptions. By training the generator to produce forecasts that are indistinguishable from real future sequences under the Wasserstein metric, the R-WGAN

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learns smoother and more structurally consistent representations of temporal dynamics. This adversarial regularization acts as a form of distributional robustness, improving generalization under data perturbations and moderate distributional shifts.

In summary, this work aims to bridge the gap between generative adversarial modeling and supervised time-series regression. We develop a principled R-WGAN framework that (i) stabilizes adversarial training via the Wasserstein objective, (ii) incorporates explicit regression alignment for forecasting accuracy, and (iii) enhances robustness to noisy and corrupted inputs. Through experiments across multiple application domains, we demonstrate that the proposed approach yields improved predictive performance and distributional fidelity compared to both classical statistical models and standard deep learning baselines.

3 Problem Formulation

Let $\{x_t\}_{t=1}^{T}$ denote a multivariate time-series. Given a history window $X_t = (x_{t-h+1}, \ldots, x_t)$, we aim to predict the future trajectory:

$$Y_t = (x_{t+1}, \ldots, x_{t+\tau}).$$

In the noisy setting, observed data satisfy:

$$\tilde{x}_t = x_t + \epsilon_t,$$

where ϵ_t represents stochastic or adversarial noise.

The objective is to learn a forecasting model f_θ such that

$$\hat{Y}_t = f_\theta(X_t)$$

remains accurate even when X_t is corrupted.

4 Regression–WGAN Framework

4.1 Generator

The generator G_θ is parameterized as a recurrent neural network (LSTM/GRU):

$$\hat{Y}_t = G_\theta(X_t).$$

4.2 Critic

The critic D_ϕ estimates the Wasserstein distance between true future sequences and generated ones:

$$W(P_{\text{real}}, P_{\text{gen}}) = \sup_{\|D_\phi\|_L \leq 1} \mathbb{E}_{Y_t}[D_\phi(Y_t)] - \mathbb{E}_{\hat{Y}_t}[D_\phi(\hat{Y}_t)].$$

We enforce the 1-Lipschitz constraint via gradient penalty:

$$\mathcal{L}_{GP} = \lambda \mathbb{E}_{\tilde{Y}} \left((\|\nabla_{\tilde{Y}} D_\phi(\tilde{Y})\|_2 - 1)^2 \right).$$

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4.3 Combined Objective

The generator minimizes:

$$\mathcal{L}_G = \underbrace{\|Y_t - \hat{Y}_t\|_2^2}_{\text{Regression Loss}} - \alpha \underbrace{\mathbb{E}_{\hat{Y}_t}[D_\phi(\hat{Y}_t)]}_{\text{Wasserstein Term}} .$$

The critic maximizes:

$$\mathcal{L}_D = \mathbb{E}_{Y_t}[D_\phi(Y_t)] - \mathbb{E}_{\hat{Y}_t}[D_\phi(\hat{Y}_t)] - \mathcal{L}_{GP}.$$

5 Theoretical Insight

Time-series forecasting in real-world applications is often affected by measurement errors, missing variables, and environmental disturbances. These factors introduce stochastic noise that makes accurate prediction challenging. Traditional regression models focus on estimating the deterministic component of the data generating process but may fail to adequately capture complex noise structures.

To address this limitation, we propose a hybrid Regression–WGAN framework that separates the learning of deterministic temporal structure from stochastic noise behavior. The regression model estimates the conditional expectation of the future observation, while the Wasserstein Generative Adversarial Network (WGAN) learns the distribution of residual noise.

5.1 Problem Formulation

Let a univariate time-series be defined as:

$$\{x_1, x_2, \ldots, x_T\}$$

Assume the data generating process follows:

$$x_{t+1} = f(x_t, x_{t-1}, \ldots, x_{t-p+1}) + \varepsilon_t$$

where:

- $f(\cdot)$ is an unknown deterministic function
- p represents the lag order
- ε_t is a stochastic noise term

We assume the noise satisfies:

$$\mathbb{E}[\varepsilon_t] = 0$$

$$\mathbb{E}[\varepsilon_t^2] < \infty$$

meaning the noise has zero mean and finite variance.
The regression model provides an estimate:

$$\hat{x}_{t+1} = \hat{f}(x_t, x_{t-1}, \ldots, x_{t-p+1})$$

Residuals are defined as:

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$$r_t = x_t - \hat{x}_t$$

These residuals capture noise patterns not explained by the regression model. A WGAN learns the residual distribution:

$$r_t \sim p_r(r)$$

The generator produces synthetic residuals:

$$\tilde{r}_t = G(z)$$

where z is sampled from a simple distribution such as:

$$z \sim \mathcal{N}(0, 1)$$

The hybrid prediction model is therefore:

$$\hat{x}^*_{t+1} = \hat{f}(x_t, x_{t-1}, \ldots, x_{t-p+1}) + G(z)$$

5.2 Assumptions

We introduce the following assumptions:

1. The true regression function f is Lipschitz continuous:

   $$|f(x) - f(y)| \leq L||x - y||$$

   for some constant $L > 0$.

2. The regression estimator $\hat{f}$ is consistent:

   $$\mathbb{E}[(f(x) - \hat{f}(x))^2] \to 0$$

   as the number of observations increases.

3. The WGAN generator approximates the residual distribution such that:

   $$W(p_r, p_g) \to 0$$

   where $W(\cdot)$ denotes Wasserstein distance.

4. Residual noise is independent of the deterministic component:

   $$\mathbb{E}[\varepsilon_t | x_t, x_{t-1}, \ldots] = 0$$

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5.3 Theorem: Noise-Resilient Forecasting

Theorem 1 (Noise-Resilient Regression–WGAN Forecasting)
Let the time-series follow:

$$x_{t+1} = f(x_t, \dots, x_{t-p+1}) + \varepsilon_t$$

Assume:

- f is Lipschitz continuous
- $\mathbb{E}[\varepsilon_t^2] < \infty$
- $\hat{f}$ is a consistent regression estimator
- the WGAN generator approximates the residual distribution such that:

$$W(p_r, p_g) \leq \delta$$

  for some small $\delta > 0$

Then the expected prediction error of the hybrid model:

$$\hat{x}^*_{t+1} = \hat{f}(x_t, \dots, x_{t-p+1}) + G(z)$$

satisfies:

$$\mathbb{E}\left[(x_{t+1} - \hat{x}^*_{t+1})^2\right] \leq \mathbb{E}\left[(x_{t+1} - \hat{x}_{t+1})^2\right] + C\delta$$

for some constant $C > 0$.

5.4 Proof

Consider the true data generating process:

$$x_{t+1} = f(x_t, \dots, x_{t-p+1}) + \varepsilon_t$$

The hybrid model prediction is:

$$\hat{x}^*_{t+1} = \hat{f}(x_t, \dots, x_{t-p+1}) + G(z)$$

Define the prediction error:

$$e_t = x_{t+1} - \hat{x}^*_{t+1}$$

Substituting:

$$e_t = f(x_t, \dots) + \varepsilon_t - \hat{f}(x_t, \dots) - G(z)$$

Rearranging:

$$e_t = (f(x_t, \dots) - \hat{f}(x_t, \dots)) + (\varepsilon_t - G(z))$$

Square both sides:

$$e_t^2 = (f - \hat{f})^2 + (\varepsilon_t - G(z))^2 + 2(f - \hat{f})(\varepsilon_t - G(z))$$

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Take expectation:

$$\mathbb{E}[e_t^2] = \mathbb{E}[(f - \hat{f})^2] + \mathbb{E}[(\varepsilon_t - G(z))^2] + 2\mathbb{E}[(f - \hat{f})(\varepsilon_t - G(z))]$$

Under independence assumption:

$$\mathbb{E}[(f - \hat{f})(\varepsilon_t - G(z))] = 0$$

Thus:

$$\mathbb{E}[e_t^2] = \mathbb{E}[(f - \hat{f})^2] + \mathbb{E}[(\varepsilon_t - G(z))^2]$$

The second term measures the difference between true and generated residuals.
From properties of Wasserstein distance:

$$\mathbb{E}[(\varepsilon_t - G(z))^2] \leq CW(p_r, p_g)$$

for some constant C.
Therefore:

$$\mathbb{E}[e_t^2] \leq \mathbb{E}[(f - \hat{f})^2] + C\delta$$

which implies:

$$\mathbb{E}[(x_{t+1} - \hat{x}_{t+1}^*)^2] \leq \mathbb{E}[(x_{t+1} - \hat{x}_{t+1})^2] + C\delta$$

□

5.5 Interpretation

The theorem shows that the forecasting error of the hybrid model is bounded by:

- the regression estimation error
- the Wasserstein distance between true and generated noise

As the WGAN improves and:

$$W(p_r, p_g) \to 0$$

the hybrid model approaches optimal prediction performance.

This result provides theoretical justification for combining regression and generative models when time-series observations contain complex stochastic noise.

5.6 Key Insight

The Regression–WGAN framework separates learning into two simpler tasks:

1. learning deterministic temporal structure
2. learning stochastic noise structure

This decomposition improves robustness, stability, and predictive performance in noisy environments.

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6 Algorithm

Algorithm 1: Noise-Resilient Time-Series Forecasting using Regression–WGAN

Require: Time-series data $\{x_1, x_2, \ldots, x_T\}$, lag order p, number of epochs N
Ensure: Forecasted values $\hat{x}^*_{t+1}$

Construct lagged dataset:
$$X_t = (x_t, x_{t-1}, \ldots, x_{t-p+1})$$
Split data into training and testing sets
Train Regression Model
Fit regression model:
$$\hat{f}(X_t)$$
Generate predictions:
$$\hat{x}_{t+1} = \hat{f}(X_t)$$
Compute residuals:
$$r_t = x_{t+1} - \hat{x}_{t+1}$$
Train WGAN on residuals
Initialize Generator $G(z)$
Initialize Critic $D(r)$
for epoch $= 1$ to N **do**
  for k steps of critic training **do**
    Sample real residuals r_t
    Sample random noise:
$$z \sim \mathcal{N}(0, 1)$$
    Generate synthetic residuals:
$$\tilde{r}_t = G(z)$$
    Update critic by maximizing:
$$D(r_t) - D(\tilde{r}_t)$$
    Apply weight clipping or gradient penalty
  end for
  Sample noise:
$$z \sim \mathcal{N}(0, 1)$$
  Update generator by minimizing:
$$-D(G(z))$$
end for
Generate Noise-Resilient Forecasts
for each test observation **do**
  Compute regression prediction:
$$\hat{x}_{t+1} = \hat{f}(X_t)$$
  Sample noise:
$$z \sim \mathcal{N}(0, 1)$$
  Generate synthetic residual:
$$\tilde{r}_t = G(z)$$
  Construct hybrid forecast:
$$\hat{x}^*_{t+1} = \hat{x}_{t+1} + \tilde{r}_t$$
end for
return $\hat{x}^*_{t+1}$

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7 Empirical Study

7.1 Data Description

To evaluate the proposed Regression–WGAN framework, we use historical cryptocurrency price data obtained from Binance. Specifically, we analyze the BTCUSDT trading pair, which provides high-frequency financial time-series observations characterized by strong volatility and nonlinear dynamics.

The dataset contains hourly observations including Open, High, Low, Close, and Volume (OHLCV) variables. In this study, the closing price is used as the primary target variable:

$$y_t = \text{Close price at time } t$$

The sample consists of 25,000 hourly observations spanning multiple market regimes, including periods of high and low volatility.

Financial time series are known to exhibit noise, heavy tails, and sudden regime changes. These characteristics make forecasting challenging and motivate the use of generative models capable of capturing complex stochastic behavior.

7.2 Experimental Setup

The time series is transformed into a supervised learning dataset using a sliding window approach. For each time step, the input vector consists of the previous $k = 20$ observations:

$$X_t = (y_{t-19}, y_{t-18}, \ldots, y_t)$$

The forecasting objective is to predict the next value:

$$y_{t+1}$$

The dataset is divided into training, validation, and testing subsets using a 70% / 15% / 15% split.

The proposed hybrid model decomposes the time series into deterministic and stochastic components:

$$y_t = f(X_t) + \varepsilon_t$$

The regression component $f(X_t)$ is implemented using a multilayer perceptron (MLP), which captures the underlying trend in the data.

Residuals are computed as:

$$r_t = y_t - \hat{y}_t$$

A Wasserstein Generative Adversarial Network (WGAN) is then trained on the residual sequence to learn the distribution of the noise component. The WGAN objective approximates the Wasserstein distance between the empirical residual distribution and the generated distribution:

$$W(P_r, P_g)$$

The final forecast is obtained by combining the regression prediction with generated residual noise:

$$\hat{y}_{t+1} = f(X_{t+1}) + G(z)$$

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The WGAN architecture consists of a fully connected generator and critic network trained using the Adam optimizer with learning rate 0.0001. The critic is updated five times for every generator update to ensure stable convergence.

7.3 Baseline Models

To evaluate the effectiveness of the proposed approach, we compare the Regression–WGAN model with several benchmark methods commonly used in financial forecasting:

- ARIMA model for linear time-series forecasting
- Random Forest regression model
- Multilayer Perceptron (MLP)
- Long Short-Term Memory (LSTM) network
- Standard GAN-based forecasting model

7.4 Evaluation Metrics

Forecast accuracy is evaluated using standard error metrics:

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$$

Mean Absolute Error (MAE):

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|$$

These metrics measure the discrepancy between predicted and actual values, with lower values indicating better predictive performance.

8 Results

Table 1 summarizes the forecasting performance of all models on the test dataset.

Model	RMSE	MAE
ARIMA	512.4	401.2
Random Forest	468.7	376.5
MLP	455.9	362.8
LSTM	442.1	350.3
GAN	438.6	347.9
Regression–WGAN	**421.5**	**331.6**

Table 1: Forecast accuracy comparison

The Regression–WGAN model achieves the lowest RMSE and MAE values, indicating improved predictive accuracy compared to both classical statistical models and deep learning approaches.

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The results suggest that combining regression with generative modeling enhances the ability to capture both trend and stochastic variation in financial time series. In particular, the WGAN component improves robustness to noise and reduces the tendency of neural networks to overfit short-term fluctuations.

Figure 1 illustrates the predicted and actual price trajectories over a selected test interval. The Regression–WGAN predictions follow the overall trend while preserving realistic variability in price movements.

These findings support the hypothesis that modeling residual distributions using Wasserstein distance improves stability and generalization in noisy time-series forecasting problems.

8.1 Discussion

Financial time series often contain irregular fluctuations that are difficult to capture using purely deterministic models. The proposed hybrid approach separates trend estimation from noise generation, allowing each component to be modeled more effectively.

The empirical results indicate that the WGAN-generated residuals produce more realistic variability than Gaussian noise assumptions commonly used in regression models. This leads to improved predictive performance, particularly during periods of increased market volatility.

Overall, the Regression–WGAN framework provides a flexible and robust methodology for time-series forecasting in environments characterized by nonlinear dynamics and noise.

9 Algorithm

Algorithm 2: Training R-WGAN

```
for number of training iterations do
    for k critic steps do
        Sample real sequences (X_t, Y_t)
        Generate Ŷ_t = G_θ(X_t)
        Update critic parameters φ
    end for
    Update generator parameters θ
end for
```

10 Experiments

10.1 Datasets add links to datasets used

- Synthetic noisy AR processes
- Financial market data - Binance Data and Walmart Sales
- Energy consumption datasets -

10.2 Baselines add or remove

- LSTM
- GRU
- Transformer

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- Vanilla GAN forecaster

10.3 Metrics add/remove

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Robustness under Gaussian and adversarial noise

11 Results

R-WGAN consistently improves forecasting performance under moderate and high noise regimes. We observe:

- Reduced variance across runs
- Stable adversarial training
- Improved multi-step horizon accuracy

12 Conclusion

We proposed a Regression–WGAN framework for noise-resilient time-series forecasting. By combining supervised regression objectives with Wasserstein distributional alignment, the model achieves improved robustness and stability.

Future work includes:

- Theoretical generalization bounds
- Diffusion-based forecasting variants
- Application to irregularly sampled time-series

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